

Benjamin Roget

Software Engineer — Backend & Systems

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SUMMARY

Backend-focused software engineer (co-op) designing reliable, maintainable systems through rigorous automated testing and declarative infrastructure (NixOS, Docker). Experienced in Java Spring Boot, Rust, and full-stack delivery. Seeking a 3-month software engineering internship in the US, starting June 2027.

EXPERIENCE

Software Engineer (Co-op) — Numih France (DOGRH)

Sep 2025 – Present

- Architected a unified E2E test engine in Java Spring Boot and Selenium, enabling identical execution via Maven in CI and an async REST API.
- Implemented cold precondition checks failing invalid runs in ~50ms (vs 20s browser startup), cutting false-positive CI failures by ~70%.
- Designed automated data-isolation routines in Jenkins pipelines to eliminate concurrent test collisions on shared environments.
- Developed and deployed a core Spring Boot internal automation tool to production.

portfolio.qgroget.com/en/blog/automated-testing-framework

Fullstack Developer Intern — Ekeep-IT

Apr 2025 – Jun 2025

- Engineered RESTful endpoints and backend services in Java Spring Boot, seamlessly integrating with relational databases.
- Developed responsive user interfaces in Next.js (React) and delivered sprint user stories in an Agile/Scrum environment.

SELECTED PROJECTS

min-btc-node

github.com/Strange500/min-btc-node

Built a Bitcoin node from scratch in Rust, implementing the P2P wire protocol (handshake, inventory, block headers) with zero protocol dependencies and concurrent peer parsing.

Rust · Bitcoin · Networking

nixos-config

portfolio.qgroget.com/en/blog/nixos-2026-reproducible

Declared multi-host NixOS configurations and dotfiles with Nix flakes and Home Manager to make development environments versioned and fully reproducible across machines.

Nix · NixOS · Linux · IaC

pixel-war

github.com/Strange500/pixel-war

Architected a Web3 pixel-canvas dApp in Solidity with pull-over-push refund patterns to prevent reentrancy attacks, and on-chain ERC-721 snapshot minting.

Web3 · Solidity · JavaScript · Blockchain

HomeLab Infrastructure

Provisioned a self-hosted server with Unraid and Docker running Jellyfin, Gitea, and Pi-hole to learn bare-metal provisioning and network routing hands-on.

Unraid · Docker · Linux · Self-hosting

RayTracer-Rust

github.com/Strange500/RayTracer-Rust

CPU ray tracer in Rust with Bounding Volume Hierarchy (BVH) spatial partitioning for O(log N) ray-primitive intersections, Phong shading, and reflections.

Rust · Graphics · Ray Tracing · BVH

QGChat

github.com/Strange500/QGChat

Full-stack chat app in Java EE with Docker Compose deployment, SQL persistence, and a JavaScript frontend.

Java · Java EE · Docker · WebSocket · SQL

ascii-cube-rs

github.com/Strange500/ascii-cube-rs

Real-time ASCII 3D cube renderer compiled to WebAssembly via a Nix flake — powers the hero widget on this portfolio.

Rust · WebAssembly · Nix

SKILLS

Languages & Systems:

Rust · Java · C · Python · TypeScript · JavaScript · WebAssembly

Backend & Web:

Spring Boot · Java EE · React · Next.js · Node.js · Angular

DevOps & Infrastructure:

NixOS · Linux · Docker · Podman · Jenkins · Traefik · Git

Architecture & Quality:

Selenium · JUnit 5 · Design Patterns · Agile

EDUCATION

Master of Engineering — Software Engineering

Sep 2025 – Present

IMT Nord Europe, Institut Mines-Télécom

Bachelor of Science — Computer Science

Sep 2023 – Jun 2025

University Institute of Technology of Lille — *Graduated top of the class*

LANGUAGES & INTERESTS

Languages: French (Native) · English (Professional — TOEIC 980/990) · Spanish (Intermediate)

Interests: Sport shooting · Running · Swimming · Cycling · Weightlifting